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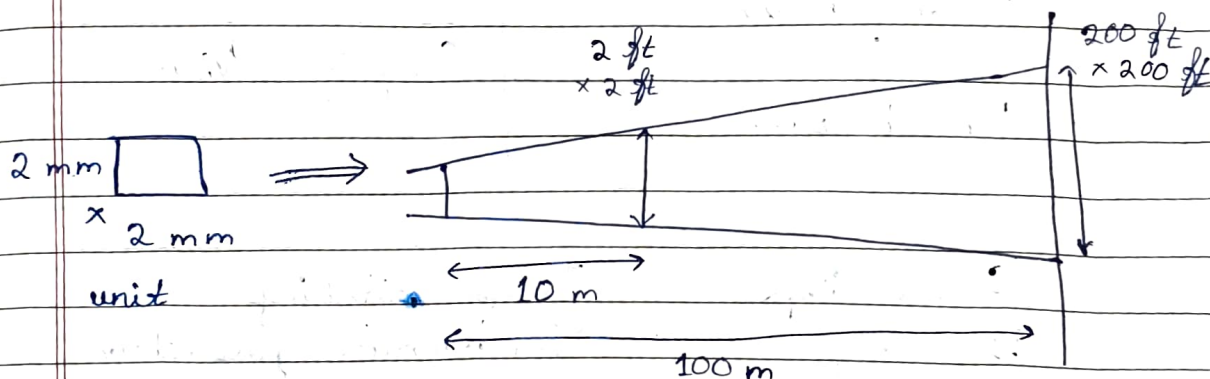
Digital Image Processing

Acquisition of

Images such as: Face Recognition, Google Maps, UAV-based photogrammetry, Instagram filters, MRI scans, satellite imagery, etc. CCTV cameras

"Earth Observation Systems"

✳ Levels of Quantization + Grid-level units + Sampling Rates



"Feature Map" generated from image information based on neuronal processing at multiple layers

Continuous light spectrum $\rightarrow$ digital image (pixelated matrix form)

Image Processing-based Object Detection Example:

Original selfie



increase brightness



Remove noise



Detect noise



Recognize person



Unlock phone

DIP

Computer Vision

Google Maps → Image stitching

Drone mapping → Orthomosaic generation

Major Recent Advancements in DIP:-

YOLO versions, CNNs, Multimodal AI, AlexNet, ResNet, Vision Transformers

$(YCbCr)$ ← YCBCR → Digital Image and Video elements

X-Ray: Bones are white



apply inverse transformations
to visualize them better
as compared to muscles and fats

DIP Pipeline:-

Illumination



Reflection ($f(x, y, \lambda)$)



Optical Lens ($O(\cdot)$)



Sensor Plane ($f(x, y)$)



Sampling ($f(i, j)$)



Quantization ($q(i, j)$)



Digital Image ($g(i, j)$)



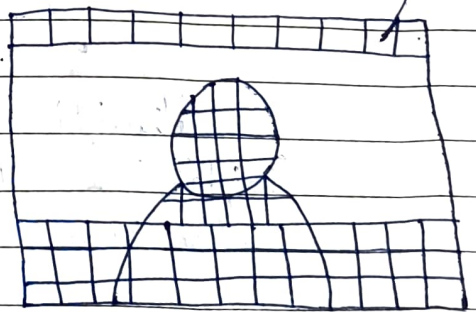
Encoding/Storage/
Transmission

CT → Computer Tomography

1 × 100

MacFarlane's images:

The box units inside the person's pixel boundaries are kept blank.



The image attainment is done using a Perforated Paper Strip.

Sir Frank MacFarlane Burnet (Nobel Laureate 1960)
in Medicine

Contrast:- Variations in the intensity of pixels

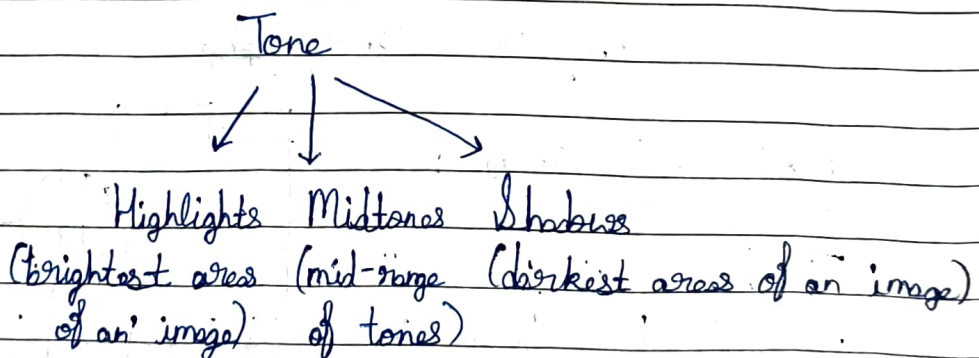
(*) Negatives: An image negative inverts the intensity levels of the original image.

Transformation: $S = 255 - r$ (for 8-bit grayscale image (0, 255))
 where r = input pixel
 S = output pixel

(*) Dispositives: A transparent, ^{positive} photograph that displays the colors and tones exactly as they appear in reality.

$S = r \rightarrow \checkmark$: (output = input pixel)

(*) Tones: The variations in brightness, contrast, and color intensity across an image.



Sir's notes: (Continued ²²/₅!)

All images discussed till Moon's surface image (in slide) are grayscale images (not RGB images)

Moon's image at that time were received at Earth

via the radio - and - microwaves (USB → Unified S-band radiowave system) ✓

(*) (E M Spectrum in slide)

(*) PET image (Position Emission Tomography)

The tracer accumulates in ~~metabolically~~ active tissues, instead of bones. metabolically

Gamma Rays के उपयोग से उत resulting bright spots को detect किया जाता है।

(*) X-Ray Imaging: Grayscale images are generated

Dense materials absorb more X-Rays → appear white

less dense materials allow more X-Rays to pass
↓
appear darker

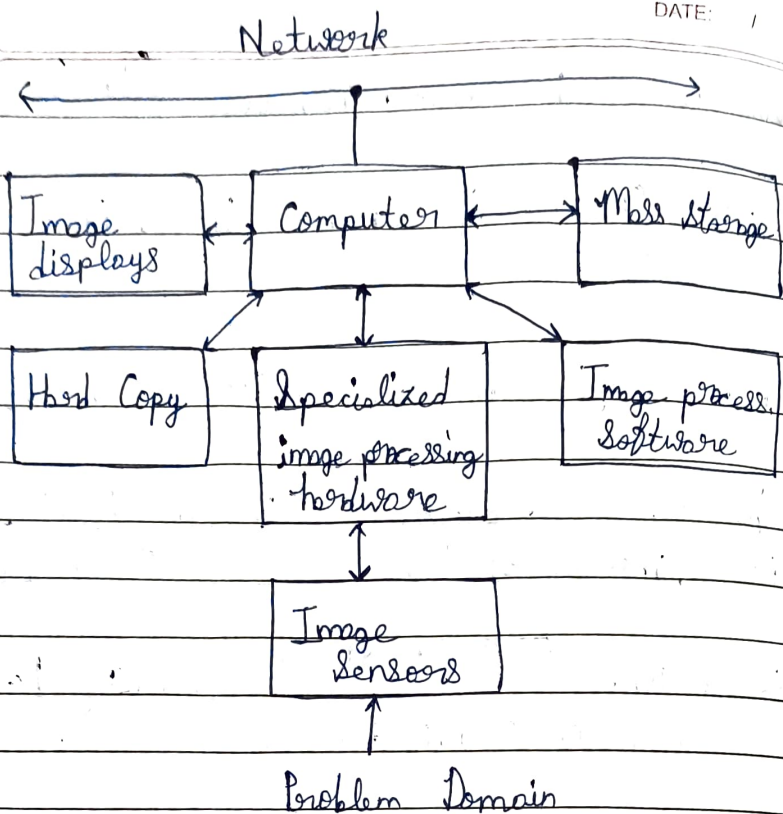
Modern X-Ray imaging:

High-pass filter,
Digital Radiography (DR)

Note: General microscopy is reflective in nature.

Light microscopy → to detect crystalline structures, such as deposition of stones and cholesterol.

("Destructive vs Non-Destructive Testing on Images")



⇒ COMPONENTS of a General Purpose Image Processing System

(*) Infrared Imaging:-

- Usually grayscale / false-color
- $\lambda \Rightarrow 0.75 \mu\text{m} - 14 \mu\text{m}$ (near IR to thermal IR)
- Represents temperature, not visible color
- Fever screening, fire detection, search and rescue, etc.

(*) Multispectral Imaging:-

- Typically: 4-13 Spectral bands
- Eg:- Blue, Green, Red, NIR, SWIR.

NIR $\rightarrow$ Near IR

SWIR $\rightarrow$ Shortwave IR

$\rightarrow$ Applications:-

Water resources, Land-use mapping
Forest monitoring, Agriculture,
Disaster management, etc.

$\rightarrow$ 20 MP > 10 MP

Resolution:- Ability to distinguish closely spaced objects.

(*) Hyperspectral Imaging:

$\rightarrow$ 100 - 300+ narrow spectral bands

$\rightarrow$ Measures detailed spectral signature

$\rightarrow$ Uses: Crop disease detection, Mineral exploration and mapping, Cancerous cell detection, etc.

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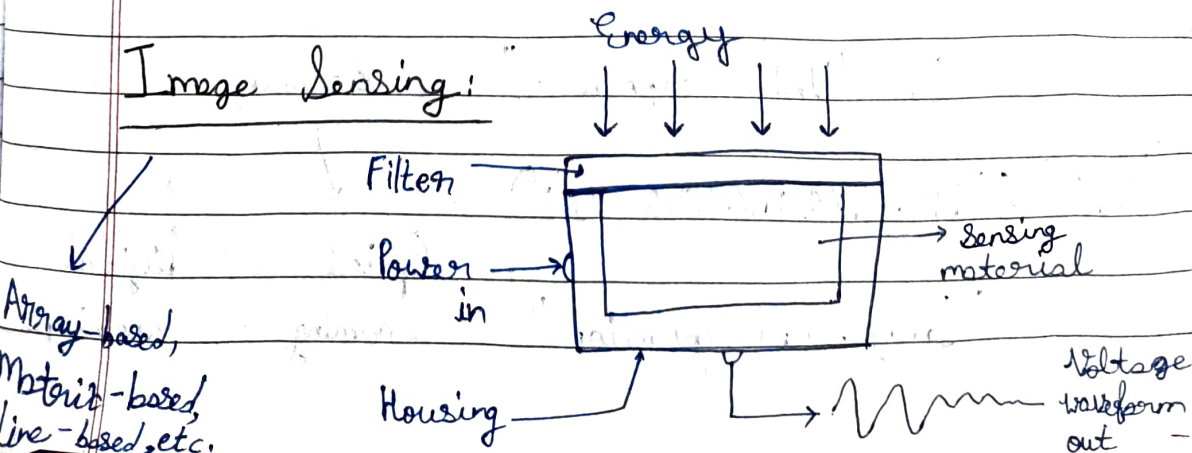
"AI camera features and one table in slide"

Human Eyes

Scotopic, Photopic, Glare limit, Brightness Order

Rods & Cones

Image Sensing:



"Please see the Helical-type sensor approach" in slide

Spatial Resolution → Measure of the smallest detail an imaging system can clearly show

Radiometric Resolution → Measure and record fine differences in light energy, expressed as bit depth (in terms of 'bits')

Image Quantization

Process of reducing the number of colors or grayscale levels used to represent an image

→ Grouping: Groups a continuous range of light intensities into a single discrete code value.

→ Compression:

→ Fewer bits per pixel reduce the overall file size, saving storage size.

→ Lossy Process: It discards visual info that can never be perfectly recovered.

"Image Denoising"

"Image Saturation"

"Over/Under-exposure"

Zooming

Image file types →

"JPEG/JPG/PNG/SVG/TIFF"

Optical
Zooming

Digital
Zooming

"Image Interpolation"

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10	20
30	40

Image Interpolation

A technique to estimate the missing pixel values and calculating new pixel grids to scale up or down smoothly in a continuous manner.

Types:-

→ Image Resizing (Upscaling / Downscaling)

i. Bilinear interpolation

image resizing technique to calculate new pixel values by average a 2×2 grid of the four closest pixels.

It performs linear interpolation first horizontally, then vertically.

10	15	20
----	----	----

It assigns mathematical weights to four surrounding pixels based on their distance from the new pixel coordinate.

Presents the blocky "Staircase" effect shown in NN interpolation.

ii. Nearest Neighbour interpolation:

In this approach, for every new pixel position, the closest corresponding pixel is located and its exact color / intensity value is copied (without averaging).

No new / intermediate colors are generated.

Enlarging an image creates harsh, visible steps or blocks (instead of smooth) because individual pixels are simply duplicated.

("Blocky" / Staircase Effect)

Tagged Edges

iii.

Bicubic Interpolation:

It calculates the average color value of 16 surrounding pixels (4×4 grid) and applies a non-linear mathematical curve (cubic spline) across the rows and columns.

A new, blended color value is generated.

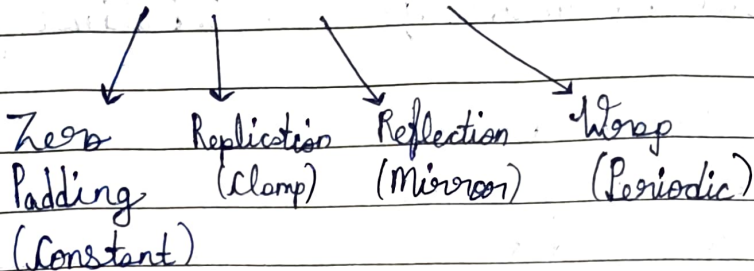
Gentler transitions, sharper details and fewer artifacts generated.

1. 2D interpolation:

$$\sin\left(\frac{\pi x}{a}\right)$$

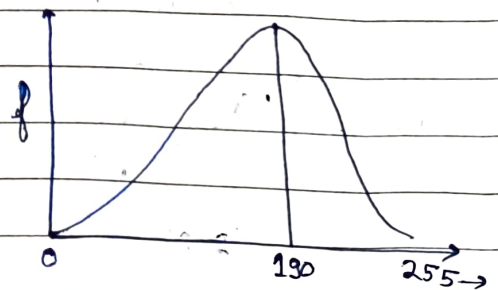
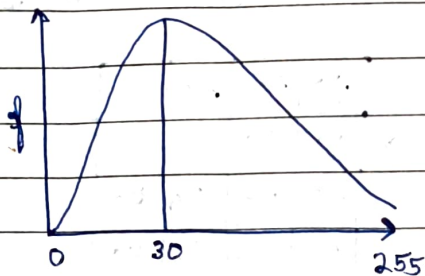
Image Padding:

It artificially expands the size of an image by adding extra pixels around its outer borders.



Pulling: Reverse - coordinate mapping process
(Inverse mapping) (Backward mapping)

The algorithm loops through every pixel on the new target canvas, maps its coordinates back to the original source image, and extracts the correct color data.



f = frequency

Image A

Image B

Ques. Which image is brighter?

Ans. Image B ✓

(because more pixels are concentrated towards the brighter (255) side, with maximum pulling at 190.)

Band Ratioing: Technique to enhance specific surface features by dividing the digital numbers (DN) values of one spectral band by another for each individual pixel.

Euclidean distance

Manhattan distance

Chessboard distance

"False color composites" $\rightarrow$ FCC

12/08/2026 यदि हम किसी 2×2 image में interpolation प्रयोग करेंगे तो image resizing करें (to 4×4)

image size will be enhanced
individual pixel size will remain the same

Eg:-

10	20
30	40

(2×2)



10	10	20	20
10	10	20	20
30	30	40	40
30	30	40	40

(4×4)

by Nearest Neighbour Interpolation

$$i_{old} = i_{new} \cdot \text{round}\left(\frac{r_1}{2}, \frac{c}{2}\right)$$

$r_1 \Rightarrow \text{row}$
 $c \Rightarrow \text{column}$
(positional values like in a matrix)

for instance,

the cell (2,2) in bigger image corresponds to old image's cell of (1,1)

$$i_{old} = i_{new} \cdot \left(\text{round}\left(\frac{r_1}{2}, \frac{c}{2}\right)\right) = 40 \cdot \text{round}\left(\frac{2}{2}, \frac{2}{2}\right)$$

$$= 40 \cdot \text{round}(1.1) \Rightarrow \therefore \text{old image cell will also be } 40.$$

If new image's cell is (3,3),
then old image's cell will be

$$i_{\text{old}} = i_{\text{new}} \cdot \text{round}\left(\frac{3}{2}, \frac{3}{2}\right)$$

$$= i_{\text{new}} \cdot (1.1)$$

$\therefore$ old image's corresponding cell will be 40.

General Formula: $i_{\text{old}} = i_{\text{new}} \cdot \text{round}\left(\frac{r}{sf}, \frac{c}{sf}\right)$

sf = scaling factor

Bilinear Interpolation:-

(above example
 $sf = 2$ ✓)

10	15	20
20	25	30
30	35	40

10	15	20
----	----	----

p_{11}	p_{12}
p_{21}	p_{22}

10	20
30	40

10	20
----	----

→ Follows the averaging approach.

p_{ij} → corresponding brightness for a cell (i,j)

Algorithm:

$$p_{\text{new}} = p_{11}(1-x)(1-y) + p_{12}(1-x)y + p_{21}x(1-y) + p_{22}xy$$

$$0 \leq x \leq 1, 0 \leq y \leq 1$$

for $x=0, y=0.5$:-

$$P_{new} = P_{11}(1-x)(1-y) + P_{12}(1-x)y + \cancel{P_{21}x(1-y)} + \cancel{P_{22}xy}$$

$$= 0.5 P_{11} + 0.5 P_{12} = \frac{P_{11}}{2} + \frac{P_{12}}{2}$$

$$\Rightarrow P_{new} = \frac{P_{11} + P_{12}}{2}$$

(= average ✓)

(*) (2×2) to (6×6) using Bilinear?

↳ 2 step process / using the Prew equation

$(2 \times 2) \Rightarrow (3 \times 3) \Rightarrow (6 \times 6)$

Note:- We can also do compression using the Bilinear approach

$$\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} \Rightarrow 25$$

Padding $\rightarrow$ Flattening

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$$\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} \Rightarrow \begin{bmatrix} 10 & - & 20 \\ - & (*) & - \\ 30 & - & 40 \end{bmatrix} \Rightarrow \begin{bmatrix} 10 & 15 & 20 \\ 20 & 25 & 30 \\ 30 & 35 & 40 \end{bmatrix}$$

$$(*) = 10 + 20 + 30 + 40 = 25$$

Bilinear Interpolation 4

(Why linear? weights of all the neighbours are equal)

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Bicubic Interpolation:

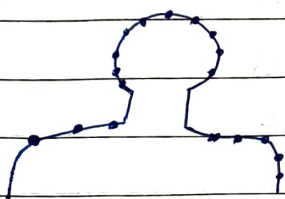
Inverse distance-based weighting approach (IDW)

Closed path $\rightarrow$ We can reach where we started
(like a circular loop) in a continuous manner

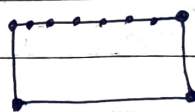
Open path $\rightarrow$ one ended path

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- Number of Corners:- (more than ⁱⁿ man-made objects, as in natural)



natural



man-made

- Circularity:- A dimensionless shape descriptor used to quantify how closely a two-dimensional binary object resembles a perfect circle.

Helps to classify different kinds of objects

$$\text{Circularity} = \frac{4\pi A}{P^2} \quad \left(\text{also known as Shape Factor or Compactness} \right)$$

$A = \text{Area}$

$P = \text{Perimeter}$

It helps to capture roughness of perimeter and overall shape form.

Circularity = 1 $\rightarrow$ perfectly smooth circle
 if 0.0 $\rightarrow$ highly elongated, irregular
 or shapes that are jagged

"Curvature" and "Radius of Curvature"

- Edge Density: - Metric to measure the concentration of edges within a specific region or across an entire image.

$$\text{Edge density} = \frac{N^2}{A}$$

N = total length of edges

A = total surface area

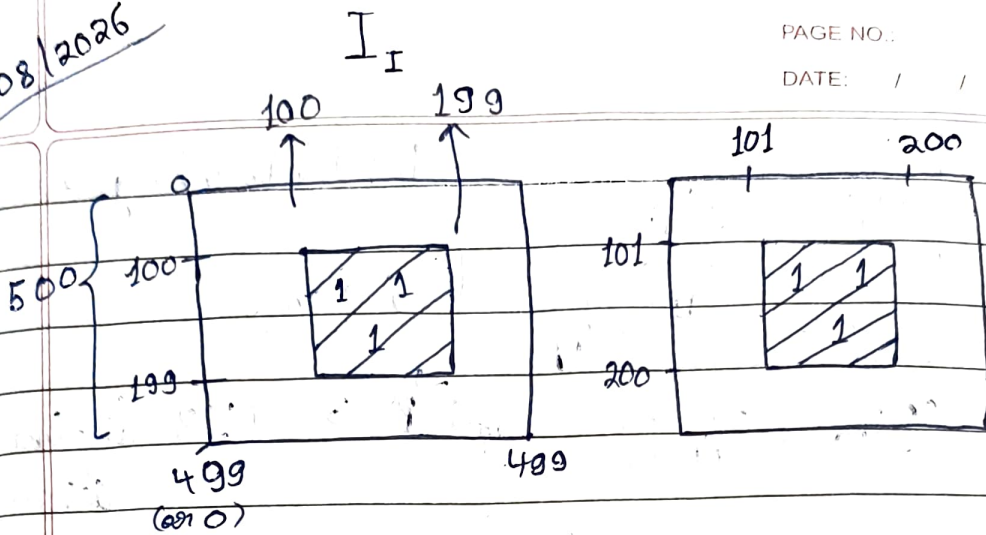
- QHull: - Quickhull algorithm to compute convex hulls across multiple dimensions (2D, 3D, or higher)

Convex hull $\rightarrow$ smallest boundary that completely encloses a set of points, shapes, or pixels, with no indentations.

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Natural Objects

Man-made objects

More no. of corners and edges relatively.

Relatively lesser no. of corners and edges

Rarely perfectly symmetric

Highly symmetrical with many parallel and perpendicular lines.

Irregular mostly and highly complex

Mostly regular and geometric in Nature with well-delineated boundaries

Fractal in Nature.

Scale-dependent.

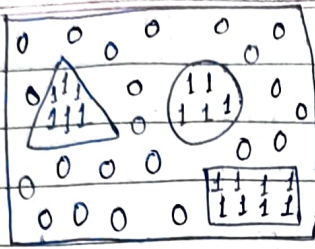
Eg: Human, Tree, Mountain, etc.

Eg:- Table, Chair, Building.

"BW Boundary" → A binary boundary / a counter, or a silhouette boundary for the perimeter of a binarized image.

Contour :- The continuous, vector-based line that traces the outer limit of a shape or object.

Eg:-



$$B = \text{BWBoundary}(i)$$

 $i \rightarrow \text{image}$

(Binary Object)

$$\text{length}(B) = 3$$

 $(\because 3 \text{ distinct objects})$


Moore's Neighbour Tracing Algorithm:-

A non-parametric technique used to find and trace the exact continuous contour of a binary object.



→ how many neighbours does this pixel?

8 ✓✓

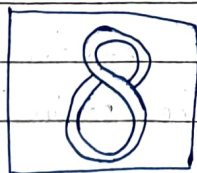
The algorithm moves clockwise around the edge pixels of an object to trace its silhouette - the boundary.

Based on the Moore Neighbourhood (8 directly surrounding pixels around a central pixel)

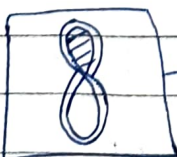
It extracts the clean boundary coordinate list.



→ how many boundaries? $\Rightarrow 2$ ✓



→ how many boundaries? $\Rightarrow 3$ ✓



→ how many boundaries? $\Rightarrow 2$

In Moore's Algorithm,

$$\theta_i = \tan^{-1} \left(\frac{\Delta y}{\Delta x} \right)$$

$d\theta \rightarrow$ derivative of that angle over time
or step distance $\left(\frac{d\theta}{ds} \right)$ ($\Rightarrow$ measures local curvature)

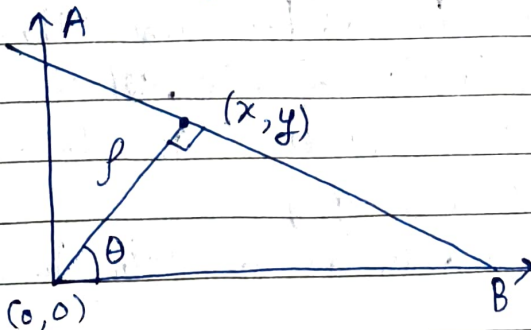
Natural: $d\theta$ is constantly changing

Man-made: $d\theta$ remains almost zero for straight path then sudden spike at sharp corners

When analyzing an edge direction θ found via the tracing path, the algorithm looks for a massive spike at $(\theta + 90^\circ)$ (or $(\theta + 180^\circ)$ / $(\theta + 270^\circ)$) across the entire contour boundary.

90° shift पर यदि सटीक shift आता रहता है अर्थात्
man-made structure identified ✓.

$$p = x \cos \theta + y \sin \theta$$



(*)

Hough Transformation:-

A technique to isolate specific shapes (straight lines and circles) within a binary (black and white) edge map.

It uses a global voting procedure in a parameter space to find geometric shapes, even if these shapes are broken, noisy, or disconnected.

Hough Transform uses Polar Coordinate System:

$$\rho = x \cos \theta + y \sin \theta$$

$\rho \rightarrow$ perpendicular distance from the origin $(0,0)$ to the line.

θ (theta) $\rightarrow$ angle of that perpendicular line relative to the horizontal axis

1. Duality: A single point (x, y) in the image space transforms into a continuous sinusoidal wave in the Hough parameter space (ρ, θ) .

2. Accumulator Array: The algorithm creates a 2D grid (matrix) of cells, where the columns represents values of θ (-90° to 90°) and rows represent values of ρ .

This matrix acts as a voting ballot.

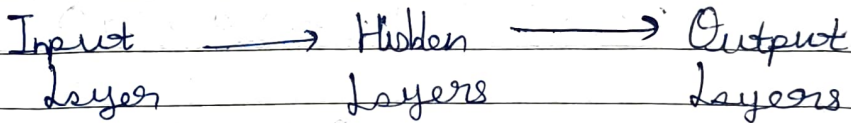
3. Voting: - The algorithm calculates all possible (ρ, θ) combinations (for every single white edge-pixel).

that could pass through that point. It adds a "+1 vote" to those corresponding cells in the accumulator array.

4. Peak detection: If many edge pixels lie along the same straight line in the image, their sinusoidal waves will all intersect at one exact point in the Hough Space.

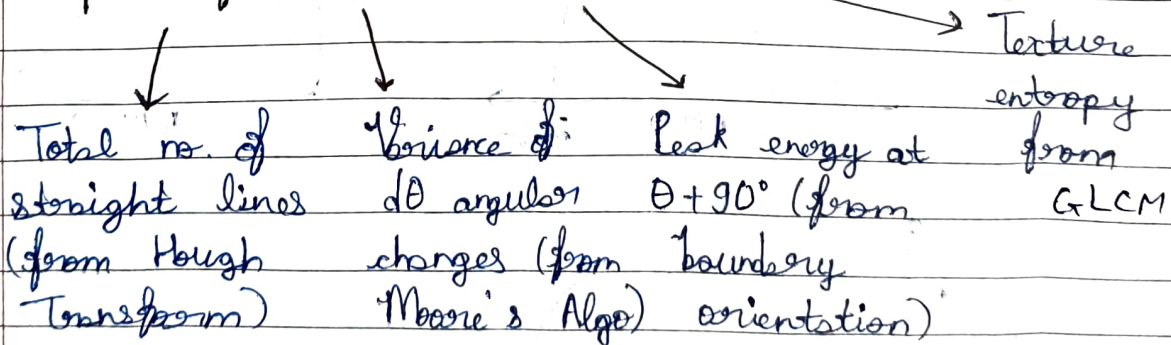
* Multi-layer Perceptron (MLP):

MLP helps to classify objects based on mathematical input features.



$$y = f(\sum (x_i \cdot w_i) + b)$$

Input features in MLP



GLCM → Gray-level co-occurrence Matrix

Note: Synthetic image is noise-free (usually) but real image will have noise inherently (Natural Image).

"Autoencoders"

"GAN → Generative Adversarial Network"

"Denoising"

does compression along with object identification (by removing redundant data)

02/09/2026

$$\text{Contrast} = \sqrt{(255-0)^2 + (\phi-0)^2 + \dots}$$

Green → 59%
Red → 29%
Blue → 11% } → sensitivity of human eye

$$\text{contrast} = \frac{255 \times 0.59 + 255 \times 0.29 + 0}{3}$$

$$= 74.8 \approx 75$$

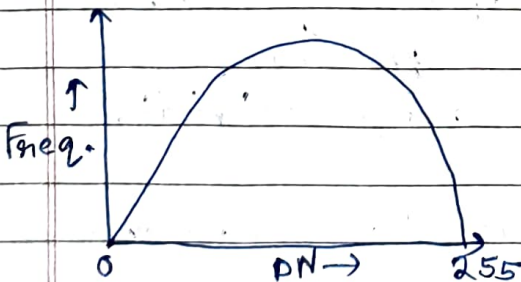
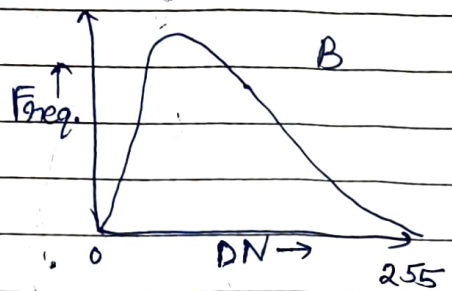
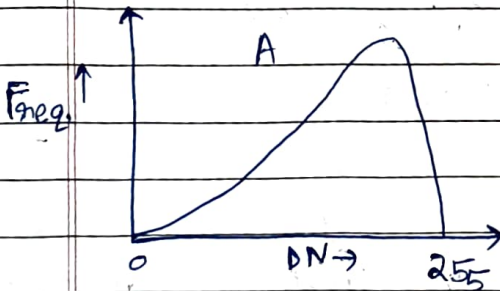


Image C has highest variance (and fact: High variance of an image is highly covered)

-loded with High Contrast of the Image).

Contrast Shifting: Image enhancement approach that adjusts the distribution of pixel intensities to change how bright and dark areas look relative to each other.

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

+5 each



6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

⇒ contrast is same
(but shifted)

Binary Thresholding:-

⇒ Intensity (brightness)
is increased

A technique to convert a grayscale image to a black-and-white binary image by comparing each pixel's intensity to a set cutoff value.

Level Slicing: Image enhancement technique that highlights a specific range of pixel intensities while dimming everything the rest.

Contrast Stretching:

An image processing technique that expands the range of pixel intensity values in an image to make dark pixels darker and bright pixels brighter.

$$B(x,y) = \frac{I_{B(x,y)} - I_{B_{min}}}{I_{B_{max}} - I_{B_{min}}} \times 255$$

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26.6

30	30	30	
30	0	30	
30	30	30	
30	30	30	
30	30	30	
30	30	30	

1	1	1
1	1	1
1	1	1

$$\begin{aligned} & (30 \times 1 + 30 \times 1 + 30 \times 1 + 30 \times 1 \\ & + 30 \times 1 + 30 \times 1 + 30 \times 1 \\ & + 30 \times 1 + 0 \times 1) / 9 \end{aligned}$$

$$= 26.6667$$

✓